

Reference basis: K. Agarwal, H. Deogun, D. Sylvester, K. Nowka, "Power Gating with Multiple Sleep Modes," *ISQED 2006*, pp. 633–637.

Design principle: The baseline replicates Agarwal et al.'s static, idle-cycle-count-only mode-selection policy — sleep decisions depend solely on consecutive idle-cycle count, with no activity-rate signal and no adaptive threshold widening. This isolates the adaptive design's actual novel contribution (the dual-condition hysteresis predictor) as the *only* variable between baseline and proposed design

Implementation approach — single-module substitution: Rather than building a second full controller from scratch, the baseline is created by replacing exactly one module:

Module	Adaptive design	Baseline	Changed?
alu_32bit.v	as-is	as-is	No
activity_monitor.v	as-is (<code>idle_count</code> used; <code>activity_rate</code> unused by baseline)	as-is	No
hysteresis_predictor.v	dual-condition, adaptive widening	replaced by <code>fixed_threshold_predictor.v</code>	Yes — this is the only change
power_state_fsm.v	as-is	as-is	No
sleep_controller.v	as-is	as-is	No
isolation_ctrl.v	as-is	as-is	No
UPF (<code>adaptive_pg.upf</code>)	as-is	as-is	No

`fixed_threshold_predictor.v` decision logic (replacing the hysteresis predictor's dual-condition gate):

- Enter Light Sleep when `idle_count ≥ LS_THRESH` (single fixed threshold, no `activity_rate` check, no enter/exit hysteresis band — exit threshold equals entry threshold).
- Enter Deep Sleep when `idle_count ≥ DS_THRESH`.
- Return to ON immediately when `idle_count` resets to 0 (`alu_active` asserted) or on `wakeup_req`.
- No `ADAPT_OFFSET` widening — thresholds never shift regardless of workload burstiness. This is the exact mechanism the Research Gap (Deliverable 1) argues is insufficient under bursty traffic, so the experiment in Step 5 will directly test that claim.
- Output encoding (`sleep_req`) identical to `hysteresis_predictor.v` so it plugs into the unmodified `power_state_fsm.v` without any other RTL changes.

Threshold values: `LS_THRESH = 3`, `DS_THRESH = 8` — same numeric values already used as *entry* thresholds in the adaptive design, so the only behavioral difference in Step 5's experiments is hysteresis/rate-awareness, not arbitrarily different threshold tuning.

Out of scope for this spec: actual Verilog implementation (that's Step 4, RTL Improvement Plan), and synthesis of the baseline (Step 5).

One optional clarification for whoever writes the actual Verilog in Step 4 (doesn't need to change this document, just flagging for later): since the baseline has no hysteresis band (`exit threshold = entry threshold`), `fixed_threshold_predictor.v` can be implemented as a purely combinational/memoryless function of `idle_count` alone — no `case` statement keyed on the current `sleep_req` state like `hysteresis_predictor.v` needs. That's a meaningful simplification worth keeping in mind when Step 4 starts, but it's an implementation detail, not a spec defect.